

MGS 602: Network Technology and Cloud Infrastructure Management

Final Lab Build a Database Server

This lab is focused on launching an Amazon RDS DB instance with high availability, configuring the DB instance to permit connections from a web server, and opening a web application to interact with the database.

Build Your DB Server and Interact With Your DB Using an App -

The lab begins with creating a security group to allow the web server to access the RDS DB instance. Then, a DB subnet group is created to specify which subnets can be used for the database. Finally, an Amazon RDS DB instance is configured and launched. The lab also covers how to wait for the database to be available and how to copy the endpoint field to use it in the web application.

This lab is to be completed on AWS Academy Portal, you can find it in under **Module 8 -**

Databases. You can begin by click Start Lab and once the environment is ready click on AWS. It will show you ways to access the lab environment. For ease of use there is an option labelled AWS SSO which will provide you with a secure url.

Note: Each lab session exists for only 2 hours, and after each session the lab gets destroyed and you have to start over again.

Background

Amazon Web Services (AWS) provides a range of database services that can be used to store, manage, and analyze data in the cloud. These services are designed to be highly scalable, durable, and secure, allowing organizations of all sizes to easily create and manage their own databases without the need for extensive infrastructure management. AWS offers a variety of database options, including both SQL and NoSQL databases, managed database services, and serverless options, to meet a wide range of use cases and applications. In this lab, we will explore how to launch and configure a Multi-AZ Amazon RDS for MySQL database instance, which provides enhanced availability and durability for production workloads.

Deliverables:

1. A written report that summarizes the key steps and outcomes of the lab exercise, including any challenges faced and lessons learned. This report should also include screenshots of the AWS Management Console and other relevant documentation.

Ans-1) The lab includes Building and Interacting with an Amazon RDS DB Instance. Below are the steps and the screenshots taken:

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The screenshot shows a web browser with multiple tabs. The active tab is a lab page from 'labs.vocareum.com'. The page title is 'Lab 5: Build Your DB Server and Interact With Your DB Using an App'. Below the title is a section 'Lab Overview and objectives'. The text describes the lab's purpose: to reinforce the concept of leveraging an AWS-managed database instance for solving relational database needs. It mentions 'Amazon Relational Database Service (Amazon RDS)' and lists supported database engines: Amazon Aurora, Oracle, Microsoft SQL Server, PostgreSQL, MySQL, and MariaDB. At the bottom, it lists objectives: launch an Amazon RDS DB instance with high availability, configure the DB instance to permit connections from a web server, and open a web application to interact with the database. The browser's taskbar at the bottom shows the Windows Start menu, search bar, and various application icons. The system tray shows the date and time as 11:48 on 22-11-2024.

Lab 5: Build Your DB Server and Interact With Your DB Using an App

Lab Overview and objectives

This lab is designed to reinforce the concept of leveraging an AWS-managed database instance for solving relational database needs.

Amazon Relational Database Service (Amazon RDS) makes it easy to set up, operate, and scale a relational database in the cloud. It provides cost-efficient and resizable capacity while managing time-consuming database administration tasks, which allows you to focus on your applications and business. Amazon RDS provides you with six familiar database engines to choose from: Amazon Aurora, Oracle, Microsoft SQL Server, PostgreSQL, MySQL and MariaDB.

By the end of this lab, you will be able to:

- Launch an Amazon RDS DB instance with high availability.
- Configure the DB instance to permit connections from your web server.
- Open a web application and interact with your database.

The screenshot shows the AWS Management Console. The left sidebar contains a navigation menu with 'Services' selected. The main content area displays the 'VPC' service page, which includes a description 'Isolated Cloud Resources', 'Top features' (Your VPCs, Subnet, Route table, Internet gateway, Egress-only internet gateways), and 'Managed Services' (IT operations management for AWS). The right sidebar shows a 'Create application' button and a search bar for applications. The bottom of the page shows the AWS logo, a search bar, and the date and time as 11:48 on 22-11-2024.

VPC

Isolated Cloud Resources

Top features

- Your VPCs
- Subnet
- Route table
- Internet gateway
- Egress-only internet gateways

Managed Services

IT operations management for AWS

Features

- Dashboard
- VPC Feature

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us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1#CreateSecurityGroup

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aws Services security group:

VPC > Security Groups > Create security group

Create security group [info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [info](#)

DB Security Group

Name cannot be edited after creation.

Description [info](#)

Permit access from Web Security Group

VPC [info](#)

vpc-04ede346beb2a7c09 (Lab VPC)

Inbound rules [info](#)

This security group has no inbound rules.

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5°C Light rain Search 11:53 22-11-2024

us-east-1.console.aws.amazon.com/vpconsole/home?region=us-east-1#CreateSecurityGroup

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aws Services security group:

N. Virginia voclabs/User3528506=Dwivedi_Divya @ 2058-1646-6154

Inbound rules [info](#)

Type info	Protocol info	Port range info	Source info	Description - optional info
MySQL/Aurora	TCP	3306	Custom sg-0c05d872b60eb0d sg-0c05d872b60eb0d1b	

Add rule

Outbound rules [info](#)

Type info	Protocol info	Port range info	Destination info	Description - optional info
All traffic	All	All	Anyw... 0.0.0.0/0	

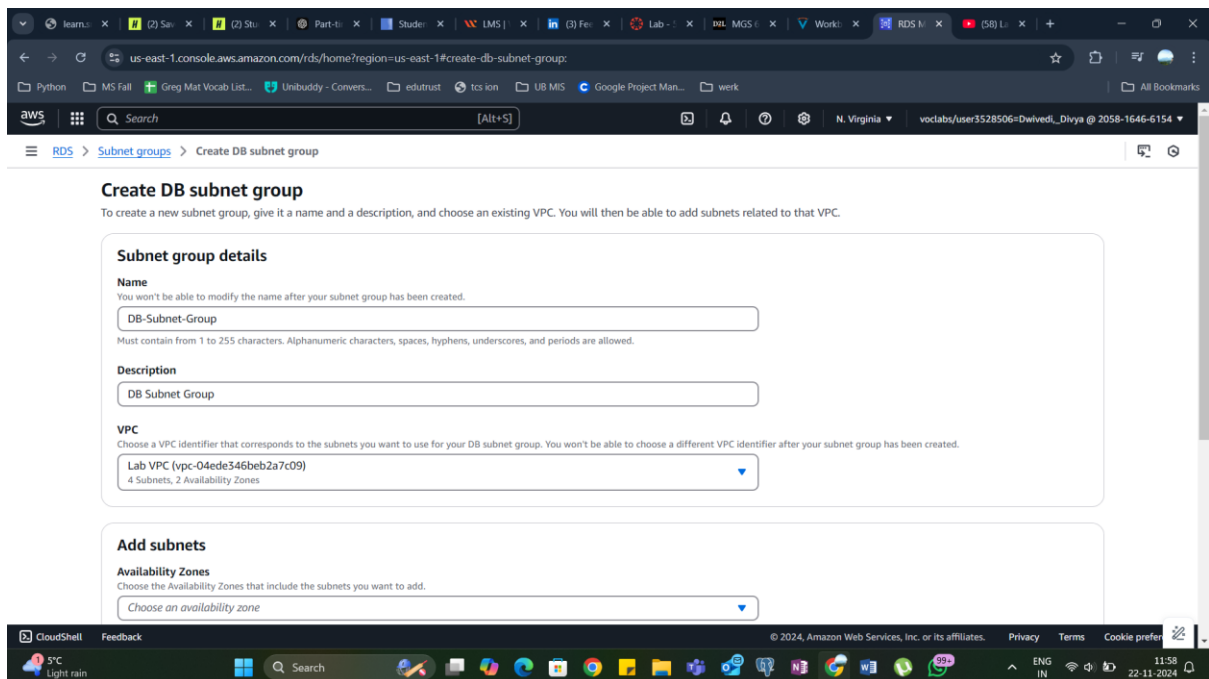
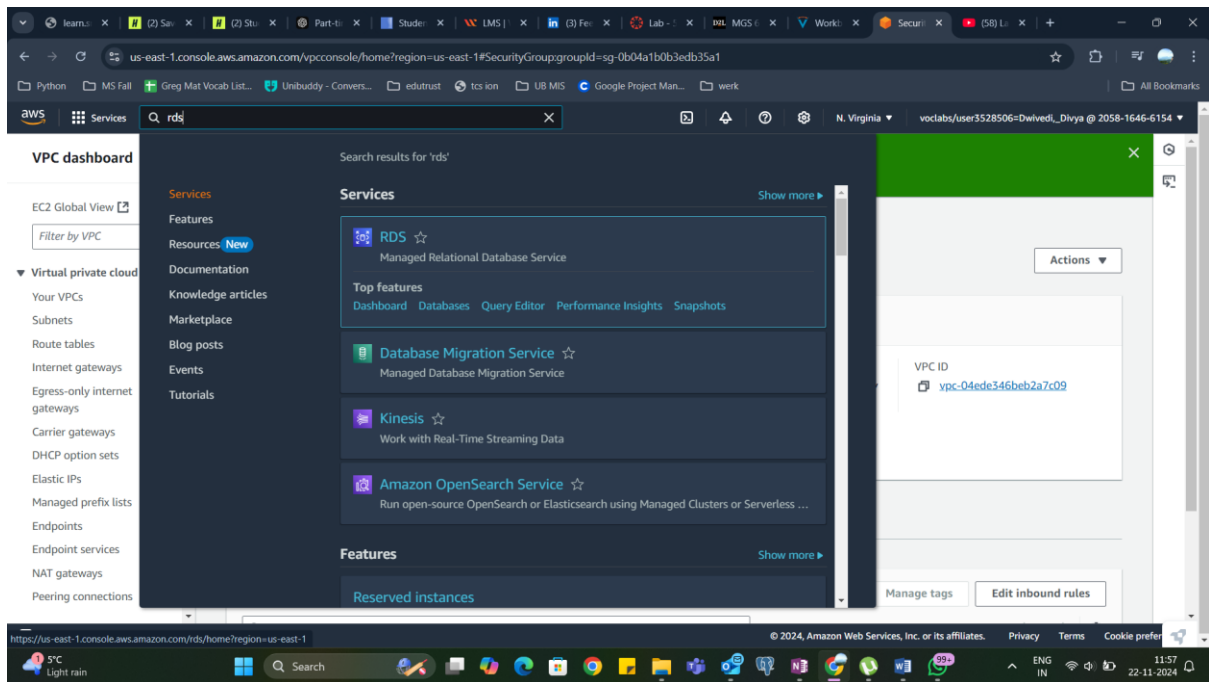
Add rule

Tags - optional

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Availability Zones
Choose the Availability Zones that include the subnets you want to add.

Choose an availability zone

us-east-1a X us-east-1b X

Subnets
Choose the subnets that you want to add. The list includes the subnets in the selected Availability Zones.

Select subnets

Private Subnet 2
Subnet ID: subnet-0cf5d0643956c4dd7 CIDR: 10.0.3.0/24

Private Subnet 1
Subnet ID: subnet-0a438b4804520d14a CIDR: 10.0.1.0/24

For Multi-AZ DB clusters, you must select 3 subnets in 3 different Availability Zones.

Subnets selected (2)

Availability zone	Subnet name	Subnet ID	CIDR block
us-east-1b	Private Subnet 2	subnet-0cf5d0643956c4dd7	10.0.3.0/24
us-east-1a	Private Subnet 1	subnet-0a438b4804520d14a	10.0.1.0/24

Create database Info

Choose a database creation method

☒ Standard create
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine options

Engine type Info

☐ Aurora (MySQL Compatible)

☒ Aurora (PostgreSQL Compatible)

☐ MySQL

☐ PostgreSQL

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The screenshot shows the AWS RDS 'Create database' console. The 'DB instance identifier' is set to 'lab-db'. Under 'Credentials Settings', the 'Master username' is 'main'. In 'Credentials management', the 'Self managed' option is selected. The 'Master password' field is masked with dots. A 'Password strength' indicator shows 'Neutral'. On the right, a 'MySQL' sidebar provides information about the database engine and its features.

Settings

DB instance identifier [Info](#)
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.
lab-db

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 60 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

▼ **Credentials Settings**

Master username [Info](#)
Type a login ID for the master user of your DB instance.
main

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - *most secure*
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed
Create your own password or have RDS create a password that you manage.

☐ Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Password strength: Neutral

MySQL [>](#)

MySQL is the most popular open source database in the world. MySQL on RDS offers the rich features of the MySQL community edition with the flexibility to easily scale compute resources or storage capacity for your database.

- Supports database size up to 64 TiB.
- Supports General Purpose, Memory Optimized, and Burstable Performance instance classes.
- Supports automated backup and point-in-time recovery.
- Supports up to 15 Read Replicas per instance, within a single Region or 5 read replicas cross-region.

The screenshot shows the AWS RDS 'Databases' console. A blue notification banner at the top states 'Creating database lab-db' and provides instructions on using settings from 'lab-db' to simplify configuration. Below the notification, a table lists the database instances. The 'lab-db' instance is shown with a status of 'Creating'. The left sidebar contains a navigation menu for various RDS features.

Amazon RDS

Dashboard
Databases
Query Editor
Performance Insights
Snapshots
Exports in Amazon S3
Automated backups
Reserved instances
Proxies

Subnet groups
Parameter groups
Option groups
Custom engine versions
Zero-ETL integrations [New](#)

Events
Event subscriptions

Creating database lab-db
Your database might take a few minutes to launch. You can use settings from lab-db to simplify configuration of suggested database additions while we finish creating your DB for you. [View credential details](#)

Databases (1)

Filter by databases

DB identifier	Status	Role	Engine	Region ...	Size	Recommendations
lab-db	Creating	Instance	MySQL Co...	us-east-1b	db.t3.micro	

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The screenshot shows the Amazon RDS console for a database instance named 'lab-db'. The status is 'Creating'. A blue notification banner at the top states: 'Creating database lab-db. Your database might take a few minutes to launch. You can use settings from lab-db to simplify configuration of suggested database add-ons while we finish creating your DB for you.' The console displays the following details:

DB identifier	Status	Role	Engine	Recommendations
lab-db	Creating	Instance	MySQL Community	
CPU	Class	Current activity	Region & AZ	
-	db.t3.micro		us-east-1b	

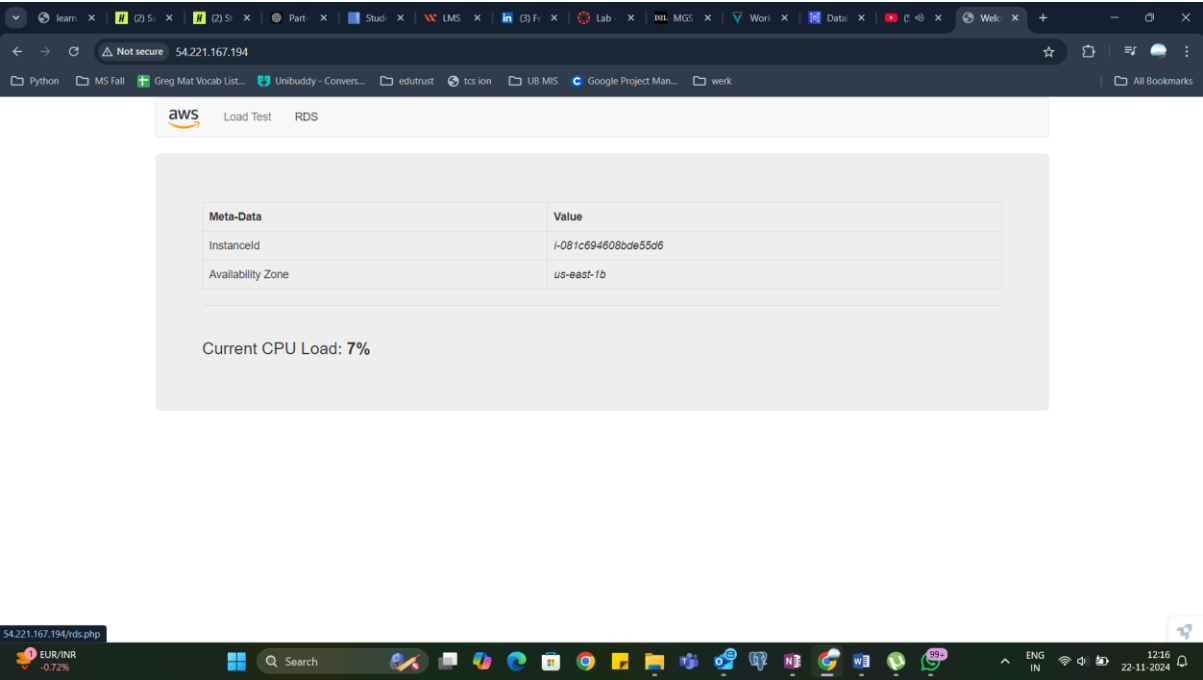
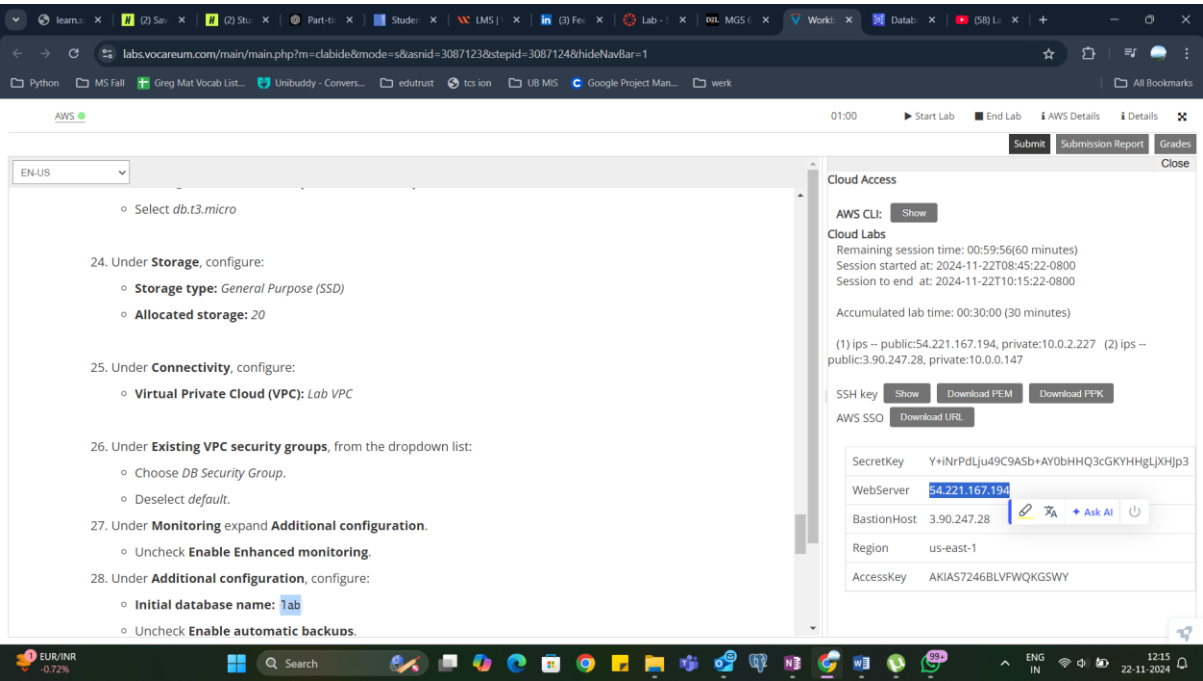
The 'Connectivity & security' tab is selected, showing the endpoint, availability zone, and VPC security groups.

The screenshot shows the Amazon RDS console for the same database instance 'lab-db'. The status is now 'Modifying'. A green notification banner at the top states: 'Successfully created database lab-db. You can use settings from lab-db to simplify configuration of suggested database add-ons while we finish creating your DB for you.' The console displays the following details:

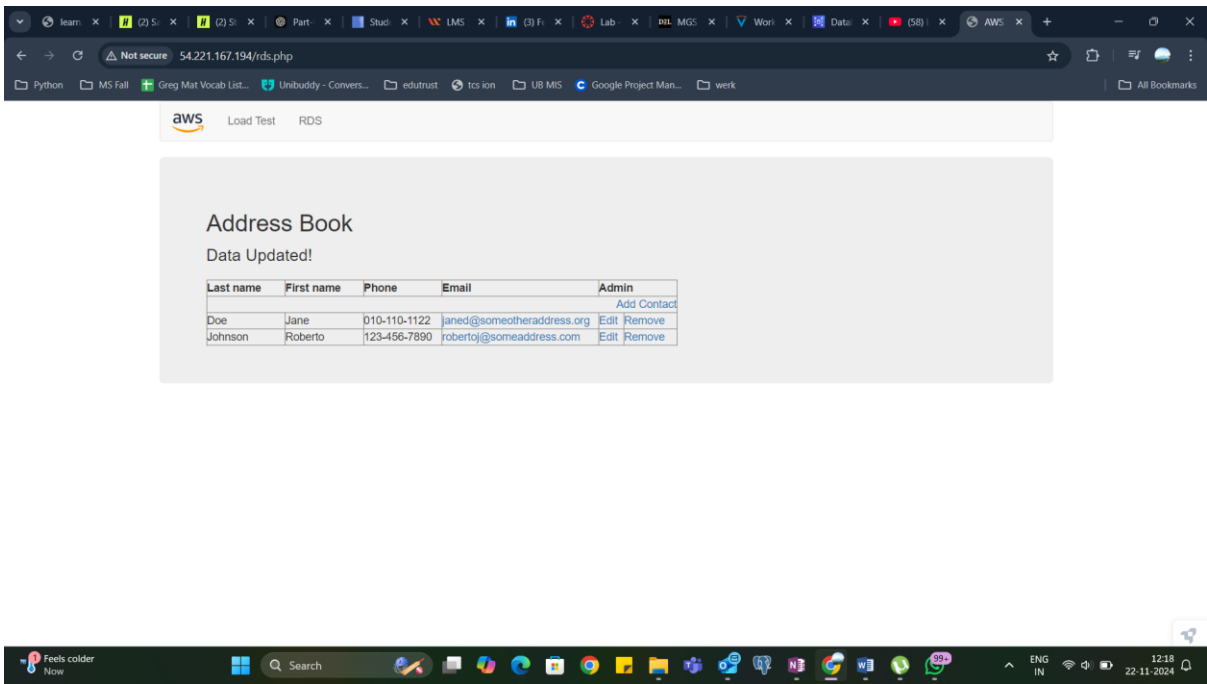
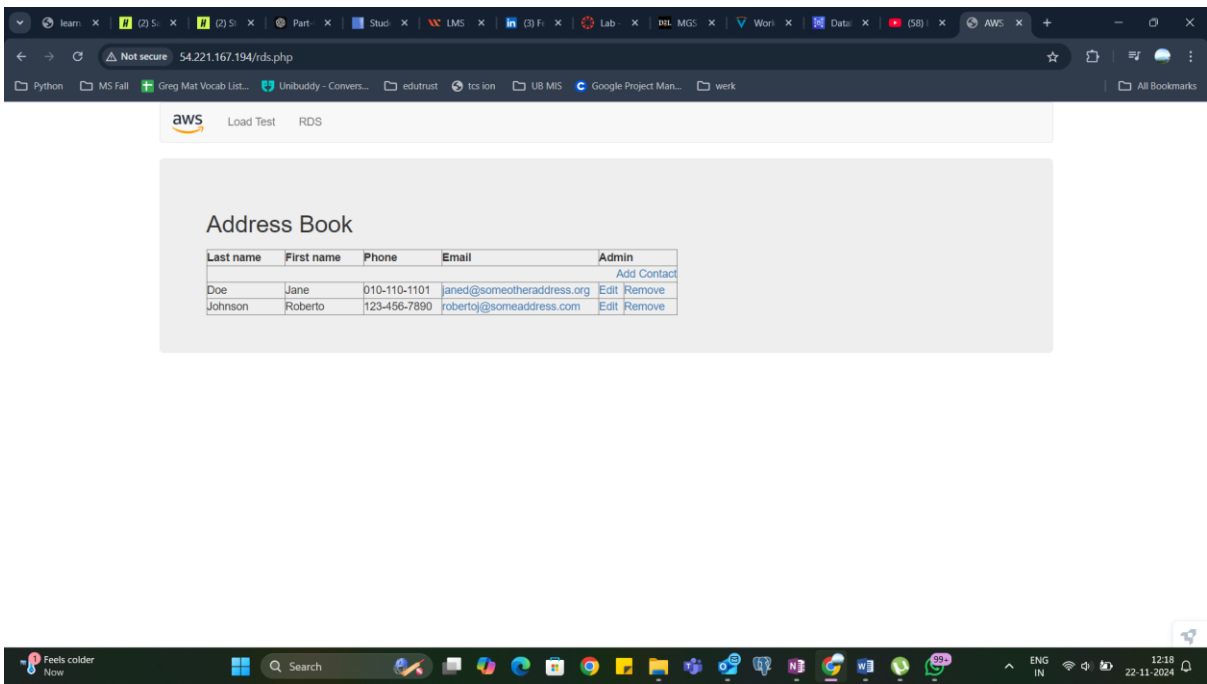
DB identifier	Status	Role	Engine	Recommendations
lab-db	Modifying	Instance	MySQL Community	
CPU	Class	Current activity	Region & AZ	
0.00%	db.t3.micro	0 Connections	us-east-1b	

The 'Connectivity & security' tab is still selected, showing the endpoint, availability zone, and VPC security groups.

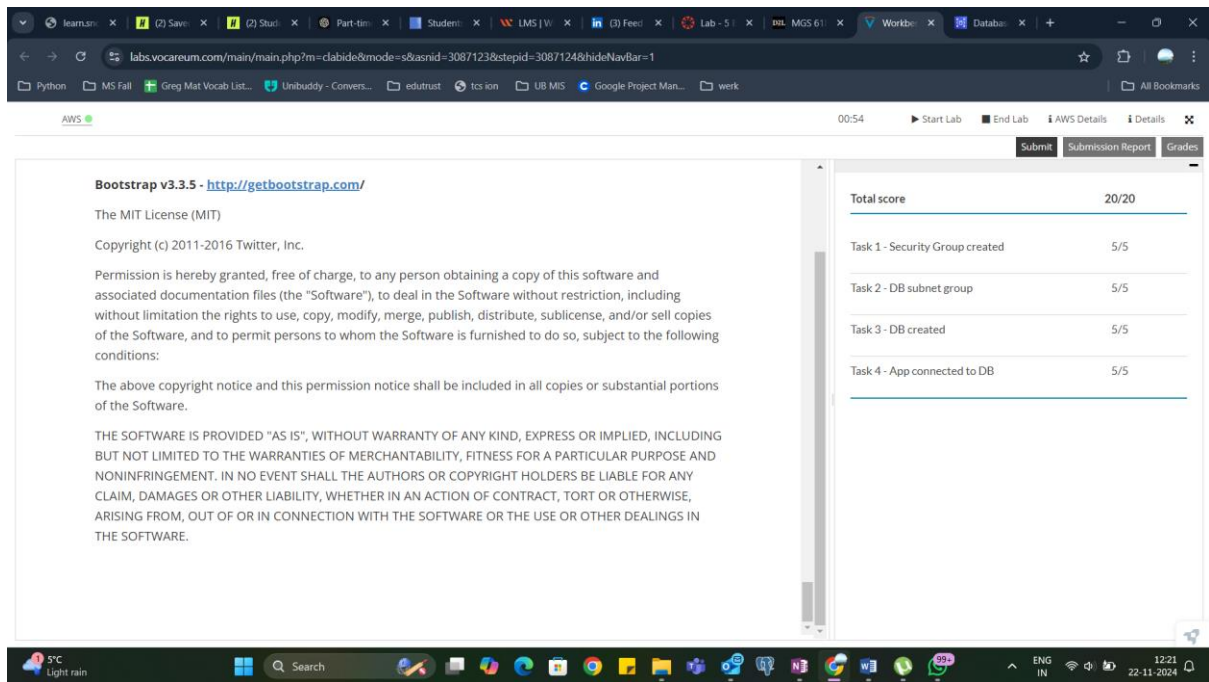
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Overall, this one was an easy lab including the creation of a Security Group, DB Subnet group, launching an Amazon RDS DB Instance, configuring connectivity, copying the end point field and interacting with the Database via a Web Application.

Some of the challenges that I faced were waiting for the database initialization and selecting the subnet. I learned that knowing and understanding how to configure security groups is very important for enabling secure and functional database access.

This lab provided a very well guided hands-on experience of configuring an Amazon RDS instance and integrating it with web application.

2. How does the multi-AZ deployment feature of Amazon RDS for MySQL database instances provide enhanced availability and durability?

Ans-2) The Multi-AZ deployment functionality duplicates data to a backup instance in a separate availability zone automatically. This duplication guarantees redundancy of data and ability to failover.

During scheduled maintenance, hardware issues, or an AZ outage, the database switches to the standby instance automatically, reducing downtime without needing manual intervention.

Synchronous replication ensures data consistency between the primary and standby instances.

3. How can you increase the security of the RDS database instance created in this lab without modifying the security group configuration?

Ans-3)

- **Activate Encryption:** Utilize AWS Key Management Service (KMS) for encrypting both the database and its backups.
- **Don't allow the general public to access the database;** only allow private IP connections through a VPN or Direct Connect.

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- IAM Database Authentication involves utilizing IAM roles and tokens for database access instead of relying on fixed passwords.
- Backup and Audit: Activate automated backups and enhance monitoring and audit logs (e.g., CloudTrail) for tracking access and changes.

4. How can you monitor the performance of the RDS database instance created in this lab, and what metrics should you be looking at?

Ans-4) Utilize Amazon CloudWatch for monitoring the RDS instance's performance. Important measurements to monitor are:

CPU Utilization refers to the proportion of CPU utilization.

Free Storage Space: Storage capacity accessible for the database.

Number of read and write operations per second is known as ReadIOPS/WriteIOPS.

Connections to the database: Count of current active database connections.

Freeable Memory refers to the quantity of memory that can be released or made available.

ReplicaLag refers to a delay in replication (when read replicas are used).

Activate Enhanced Monitoring to capture live operating system data and detailed analytics.

5. What are some key factors to consider when designing a highly available and fault-tolerant architecture in AWS? How would you approach this design process and what AWS services would you utilize?

Ans-5) Things to think about:

Avoid single point of failure by implementing multi-AZ deployments and replication.

Distribute the load by setting up Elastic Load Balancers (ELBs).

Scalability: Utilize auto-scaling groups to manage instances efficiently.

Disaster Recovery: Make sure there are backups and a plan for recovery.

Security measures such as IAM roles, security groups, and encryption should be implemented.

Methodology for the Design Process:

Recognize important elements of the application.

Create redundancy and failover mechanisms.

Utilize managed services such as RDS, DynamoDB, or S3 that offer durability capabilities.

Regularly monitor and test the architecture.

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AWS Services to Make Use Of:

RDS is used for deploying databases across multiple availability zones.

Traffic distribution using Elastic Load Balancing.

Automated scaling for adjusting computing resources.

Using CloudWatch and CloudTrail to monitor and audit the system.

Using S3 and Glacier for storing backup data.

6. In Task 1, you created a security group for your RDS DB instance to allow access from the Web Security Group. What other options do you have for controlling access to your RDS DB instance, and what are the advantages of using these options?

Ans-6) VPC Subnet Restriction: Ensure the RDS instance is located in a private subnet within a VPC, preventing direct internet access.

Benefits: Reduces vulnerability to outside risks.

IAM Database Authentication utilizes IAM roles and temporary tokens to manage access.

Benefit: Does away with the necessity of using fixed passwords.

AWS Secrets Manager allows you to securely store database credentials and retrieve them through code.

Benefit: Streamlines credential handling and improves safety measures.

Network Access Control Lists (NACLs) offer an extra level of network filtering at the subnet level.

Benefit: Grants precise management of traffic movement.

Database Firewall Rules: Utilize database security functionalities to manage IP or user role access.

Benefit: Enhances control over access to databases.